E-commerce Sales & Customer Analytics — Portfolio Case Study

Power BI Portfolio Project

1. Project Overview

This project is an E-commerce Sales and Customer Analytics dashboard developed to analyze sales performance and provide actionable business insights.

The project uses transactional sales data to examine revenue, orders, units sold, customers, products, categories, and regional performance.

The analysis was developed in Power BI, using data preparation, data modeling, DAX measures, and interactive visualizations to transform raw transactional data into a business intelligence solution.

The final dashboard provides management with a consolidated view of sales performance and enables users to explore business results across different products, customers, categories, regions, and time periods.

2. Business Problem

The business had transactional sales data but lacked a consolidated, interactive way to understand its sales performance.

Management needed to answer questions such as:

- How much revenue is being generated?
- How are sales changing over time?
- Which products and categories generate the most revenue?
- Which regions perform differently?
- How many orders and units are being sold?
- How concentrated is revenue across customers?

The objective of the project was therefore to transform the underlying sales data into an interactive Power BI business intelligence dashboard that would allow management to monitor performance, identify patterns, and investigate areas requiring further attention.

3. Dataset

The project used row-level e-commerce transactional data containing information about orders, products, customers, regions, quantities, unit prices, and sales amounts.

Working with transaction-level data allowed the analysis to calculate and compare business metrics dynamically across product, customer, category, region, and time dimensions.

4. Data Cleaning & Preparation

The raw dataset was prepared in Power Query before being loaded into the Power BI data model.

Key preparation steps included:

- Trimming whitespace from customer names.
- Standardizing region values using consistent capitalization.
- Standardizing category values.
- Identifying and removing records with missing UnitPrice values.

- Assigning appropriate data types to dates, quantities, prices, and revenue fields.
- Validating the relationship between quantity, unit price, and sales amount.
- Creating additional analytical fields such as year, month name, and line revenue.
- Splitting customer names into first and last names where required.

These steps established a consistent analytical dataset suitable for Power BI modeling and reporting.

5. Data Model

The Power BI solution used a star-schema data model designed to support flexible and reliable analysis.

The central FactSales table contained the transactional sales records, while supporting dimension tables provided descriptive information for analysis.

The model included:

- FactSales
- DimDate
- DimProduct
- DimCustomer
- DimRegion

Relationships between the fact and dimension tables enabled the dashboard to analyze sales performance across different products, customers, regions, and time periods.

This structure also supported the development of reusable DAX measures and interactive filtering throughout the dashboard.

6. Key Metrics & Analysis

Several DAX measures were developed to provide a consistent view of e-commerce performance.

Primary KPIs:

- Total Revenue: ■355.25M
- Total Orders: approximately 3K
- Total Units Sold: approximately 12K
- Total Customers: 997
- Average Order Value (AOV): approximately ■118.42K

Additional measures were developed to analyze revenue per customer, revenue per unit, customer revenue contribution, and top-customer revenue concentration.

These measures supported analysis across products, categories, customers, regions, and time.

7. Key Business Insights

1. Electronics was the leading revenue category.

Electronics generated approximately ■154.19M, representing about 43.4% of total revenue.

2. Customer revenue was relatively dispersed.

The top 10 customers contributed approximately 2.79% of total revenue, indicating that overall revenue was not highly concentrated among a small number of customers.

3. Regional performance varied.

The South region recorded the highest total revenue and revenue per customer, while the East recorded the lowest revenue per customer. These differences warrant further investigation into factors such as product mix, customer behavior, and sales practices.

4. Sales volume did not always correspond to revenue value.

Products such as Phone Stand demonstrated that high unit volume does not necessarily produce high revenue, while products such as Smart Watch combined relatively strong volume with strong revenue performance.

5. Revenue per unit varied across products.

Air Fryer had a higher revenue per unit than Wireless Earbuds. However, because cost information was not available, this metric should not be interpreted as profit margin.

8. Business Recommendations

Based on the findings, the following actions are recommended:

- Investigate the drivers of Electronics' strong revenue performance and identify opportunities to sustain or expand that performance.
- Investigate lower-performing categories, particularly Beauty, to understand whether differences are related to product demand, pricing, customer behavior, or product mix.
- Investigate regional differences, particularly the lower revenue per customer observed in the East, before making operational decisions.
- Examine product-level volume versus revenue to identify products that generate strong sales volume but relatively lower monetary value.
- Consider collecting cost and margin data so future analysis can evaluate profitability rather than revenue alone.
- Continue monitoring customer concentration, since the relatively low contribution from the top 10 customers indicates that revenue is distributed across a broad customer base.

9. Tools & Skills

- Power BI
- Power Query
- DAX
- Data cleaning and transformation
- Star-schema data modeling
- KPI development
- Interactive dashboard design
- Business intelligence analysis
- Data storytelling and insight generation
- Evidence-based business recommendations

10. Project Outcome & Limitations

The project transformed raw e-commerce transaction data into an interactive Power BI solution that enables management to analyze sales performance and investigate business opportunities.

Limitations:

- The analysis describes revenue performance but does not establish profitability because cost data was unavailable.

- Regional and product differences identify areas for investigation but do not establish causal explanations.
- The dataset supports descriptive and diagnostic analysis; additional data would be required for stronger predictive or profitability analysis.